

GOVERNMENT ENGINEERING COLLEGE, SHEIKHPURA

B.Tech second semester mid term Examination 2026

Branch: CSE, EE & ECE

Full marks: 20

Paper: Engineering Chemistry

Time: 2.0 hrs

Instructions:

- i. The marks are indicated in the right-hand margin.
- ii. There are 2 Sections in this Paper.
- iii. Attempt both Sections.

Section: - A

It contains 7 questions in which each question containing 1 mark.
Answer any 5 questions. 1 x 5 = 5 marks.

1. According to Planck's quantum theory, the energy of radiation is given by: CO1/L1
(A) $E = mc^2$ (B) $E = h\nu$ (C) $E = \frac{1}{2} m\nu^2$ (D) $E = PV$
2. Bonding molecular orbitals are formed by: CO1/L2
(A) Subtractive overlap of atomic orbitals
(B) Additive overlap of atomic orbitals
(C) No overlap of atomic orbitals
(D) Destructive interference only
3. What is the coordination number of cobalt in $[\text{Co}(\text{NH}_3)_6]^{3+}$? CO1/L1
(A) 3 (B) 4 (C) 6 (D) 2
4. Which of the following transitions is commonly observed in UV-Visible spectroscopy? CO2/L2
(A) $\sigma \rightarrow \sigma^*$ (B) $\pi \rightarrow \pi^*$ (C) $n \rightarrow \pi^*$ (D) All
5. In a lead-acid battery during discharge, which of the following reactions occurs at the anode? CO3/L2
(A) $\text{PbO}_2 \rightarrow \text{Pb}^{2+}$ (B) $\text{Pb} \rightarrow \text{PbSO}_4$
(C) $\text{SO}_4^{2-} \rightarrow \text{SO}_2$ (D) $\text{H}_2\text{SO}_4 \rightarrow \text{H}_2 + \text{SO}_4^{2-}$
6. Which of the following boiler problems is caused by the formation of hard deposits on the inner walls? CO4/L2
(A) Priming (B) Foaming
(C) Scale formation (D) Caustic embrittlement

1.100
0.591
1.691

7. Which type of transition is responsible for absorption in infrared spectroscopy? CO2/L2
- (A) Electronic transition (B) Rotational transition
(C) Vibrational transition (D) Nuclear spin transition

Section: - B

It contains 5 questions in which each question containing 5 marks.

Answer any 3 questions.

3x5 = 15 marks.

8. (a) Using Bent's Rule, explain the variation in bond angles in PCl_3F_2 . CO1/L2.
(b) Discuss structure and synthesis of metal carbonyl complexes. CO1/L2.
9. (a) Describe the photoelectric effect and derive the photoelectric equation on the basis of Einstein's theory. CO1/L3
(b) Light of frequency $1.2 \times 10^{15} \text{ Hz}$ is incident on a metal surface whose work function is $4.5 \times 10^{-19} \text{ J}$. Using Einstein Photoelectric Equation, calculate the maximum kinetic energy of the emitted electrons. CO1/L3.
10. (a) Define chemical shift in Nuclear Magnetic Resonance. CO2/L1
(b) In an NMR experiment, a proton signal appears at 300 Hz downfield from TMS using a spectrometer operating at 60 MHz. Calculate the chemical shift in ppm. CO2/L3
11. (a) State selection rules for rotational and vibrational transitions. CO2/L1
(b) A hydrogen fluoride (HF) molecule has bond force constant $k = 9.0 \times 10^2 \text{ N m}^{-1}$. The atomic masses are : $\text{H} = 1.0 \times 10^{-27} \text{ kg}$, $\text{F} = 3.15 \times 10^{-26} \text{ kg}$. Using Hooke's Law Model in IR Spectroscopy, calculate the vibrational frequency (ν) of HF molecule. CO2/L3
12. (a) Define corrosion, List any two types of corrosion. CO3/L1
(b) The standard electrode potential of the cell is $E_{\text{cell}}^\circ = 1.10 \text{ V}$. At 298 K, the concentrations are: $[\text{Zn}^{2+}] = 0.01 \text{ M}$ and $[\text{Cu}^{2+}] = 1.0 \text{ M}$. Using Nernst Equation, calculate the cell potential (E_{cell}) at non-standard conditions. CO3/L3

$$6.626 \times 10^{-34} \times 1.2 \times 10^{15} - 4.5 \times 10^{-19}$$